

DSA topic order: scratch to advanced

Java track · FAANG interview prep

Foundations

1. Big-O notation / complexity analysis
2. Arrays & Strings (traversal, in-place ops, prefix sums)
3. Hashing (HashMap/HashSet, when to use)
4. Recursion basics (base cases, call stack)
5. Linked Lists (singly, doubly, cycle detection)
6. Stacks & Queues

Core patterns

7. Two pointers
8. Sliding window
9. Fast & slow pointers
10. Binary search (+ binary search on answer)
11. Sorting-based tricks & merge intervals
12. Trees (traversals: inorder/preorder/postorder, BST properties)
13. BFS & DFS (trees, then graphs)

Intermediate / advanced structures

14. Graphs (topological sort, union-find, Dijkstra, cycle detection)
15. Heaps / priority queues
16. Tries

The big one — dynamic programming

17. 1D DP (climbing stairs, house robber, coin change)
18. 2D DP (knapsack, longest common subsequence, edit distance)
19. DP on trees/graphs (advanced, later-stage)

Advanced technique

20. Backtracking (subsets, permutations, N-queens)
21. Monotonic stack/queue

22. Bit manipulation (light touch – rarely the focus)

Why this order matters

- Don't move to the next topic until you can explain *why* a technique works, not just recite it — that's what makes pattern recognition transfer to unseen problems.
- Trees → Graphs → DP is intentionally sequential: tree traversal is the foundation for graph BFS/DFS, and DP problems are often modeled as graphs/trees under the hood.
- DP is placed after graphs on purpose — it clicks much faster once you're fluent with recursion trees and graph traversal, since DP is essentially "recursion with memory."